

## L8: GN Benchmark

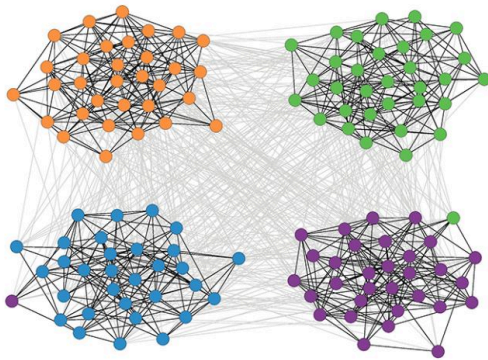


Figure 9.25 from *Network Science* by Albert-László Barabási

How can we test if a community detection algorithm can identify the right set of communities? In most real datasets and problems, we do not have such “**ground truth**” information.

So, a first task is to create synthetically generated networks with known community structure – and then to check whether a community detection algorithm can discover that structure.

Sometimes these synthetic networks are referred to as “**benchmarks**”.

The simplest benchmark is referred to as “**Girvan-Newman (GN)**”. In this model, all communities have the same size. Additionally, any pair of nodes within the same community has the same connection probability  $p_{int}$ , and any pair of nodes in two different communities also has the same connection probability  $p_{ext}$ . Consequently, the GN benchmark cannot generate a heavy-tailed degree distribution.

In more detail, we specify the number of communities  $n_c$  and the number of nodes in each community  $N_c$ .

The total number of nodes is  $N = n_c N_c$ .

We also need to specify the following parameter  $\mu$ :

$$\mu = \frac{k^{ext}}{k^{ext} + k^{int}}$$

where  $k^{int}$  is the average number of internal connections in a community and  $k^{ext}$  is the average number of external connections between communities.

If  $\mu$  is close to 0 almost all connections are internal and the community is very clearly separated.

As  $\mu$  increases, the community structure becomes weaker. After  $\mu$  is greater than 0.5, it is more likely to see external connections than internal, meaning that there is no community structure.

The visualization shows a GN network with 128 nodes, four communities, and 32 nodes per community when  $\mu$  is 40%.